Polar Equations and Graphs

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Recognizing polar graphs**

Identify the graph of each polar equation:

a $r = 3$ Circle of radius 3 centered at the pole.

b $\theta = \frac{\pi}{6}$ Line through the pole at angle $\frac{\pi}{6}$.

c $r = 2 + 2\cos\theta$ Cardioid (heart-shaped, symmetric about the polar axis).

d $r = 4\sin 3\theta$ Rose curve with 3 petals.

(a) Circle of radius 3 centered at the pole.; (b) Line through the pole at angle $\frac{\pi}{6}$.; (c) Cardioid (heart-shaped, symmetric about the polar axis).; (d) Rose curve with 3 petals.

2 **[Foundational]** • **Target: Recognizing polar graphs**

For the limaçon $r = 1 + 2\cos\theta$:

a Explain how you know it has an inner loop. In $r = a + b\cos\theta$, $|a| < |b|$ (here $1 < 2$) produces an inner loop.

b Find the values of θ in $[0, 2\pi)$ at which the curve passes through the pole. $1 + 2\cos\theta = 0$ gives $\cos\theta = -\frac{1}{2}$: $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$.

(a) In $r = a + b\cos\theta$, $|a| < |b|$ (here $1 < 2$) produces an inner loop.; (b) $1 + 2\cos\theta = 0$ gives $\cos\theta = -\frac{1}{2}$: $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$.

3 **[Foundational]** • **Target: Recognizing polar graphs**

For the rose $r = 5\sin 2\theta$:

a State the number of petals. $n = 2$ is even, so $2n = 4$ petals.

b State the maximum value of $|r|$ and the smallest positive θ at which it occurs. Maximum $|r| = 5$, first at $2\theta = \frac{\pi}{2}$, i.e. $\theta = \frac{\pi}{4}$.

(a) $n = 2$ is even, so $2n = 4$ petals.; (b) Maximum $|r| = 5$, first at $2\theta = \frac{\pi}{2}$, i.e. $\theta = \frac{\pi}{4}$.

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4 [Foundational] • Target: Recognizing polar graphs

Determine whether the graph of $r = 2 + \sin\theta$ passes through the pole, and state the type of limaçon.

$2 + \sin\theta \geq 1 > 0$ for all θ, so it never passes through the pole. Since $|a| > |b|$ $(2 > 1)$, it is a dimpled limaçon with no inner loop.

5 [Foundational] • Target: Symmetry tests

Test $r = 3\cos 2\theta$ for symmetry about the polar axis (replace θ with $-\theta$) and about the pole (replace r with $-r$).

Polar-axis: $\cos(2(-\theta)) = \cos(2\theta)$, unchanged — symmetric about the polar axis. Pole: replacing $r \to -r$ gives $-r = 3\cos 2\theta$, not equivalent to the original in general — the standard test is inconclusive for pole symmetry, though the curve (a 4-petal rose) is in fact symmetric about the pole by direct graphing.

6 [Foundational] • Target: Symmetry tests

Test $r = 2 + 2\sin\theta$ for symmetry about the line $\theta = \frac{\pi}{2}$ (replace θ with $\pi - \theta$).

$\sin(\pi - \theta) = \sin\theta$, so the equation is unchanged: symmetric about $\theta = \frac{\pi}{2}$ (the vertical axis).

7 [Foundational] • Target: Graphing from a table of values

For the cardioid $r = 1 + \cos\theta$, complete the table of r-values: $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.

a $r(0) \ = 2$ **b** $r\left(\frac{\pi}{2}\right) \ = 1$ **c** $r(\pi) \ = 0$ **d** $r\left(\frac{3\pi}{2}\right) \ = 1$

(a) $= 2$; (b) $= 1$; (c) $= 0$; (d) $= 1$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Graphing from a table of values

For the circle $r = 4\sin\theta$, find the value(s) of θ in $[0, \pi)$ where r reaches its maximum, and state that maximum value.

Maximum of $\sin\theta$ is 1 at $\theta = \frac{\pi}{2}$, giving $r = 4$.

9 [Intermediate] • Target: Rose curves in depth

For the rose $r = 3\cos 5\theta$, state the number of petals and the length of each petal (maximum $|r|$).

$n = 5$ is odd, so there are $n = 5$ petals, each of length $|r|_{\max} = 3$.

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10 [Intermediate] • Target: Rose curves in depth

Explain, using the general rule for rose curves $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$, why n odd gives n petals but n even gives $2n$ petals.

When n is odd, tracing θ from 0 to 2π retraces each petal a second time (due to negative r values overlapping earlier petals), so only n distinct petals appear. When n is even, no such overlap occurs, and all $2n$ petals from the full $[0, 2\pi)$ sweep are distinct.

11 [Intermediate] • Target: Identifying limaçon subtypes

Classify each limaçon $r = a + b\cos\theta$ as having an inner loop, being dimpled, or being convex (roughly oval, no dimple), based on the ratio $\frac{a}{b}$:

a $r = 1 + 3\cos\theta$ $\frac{a}{b} = \frac{1}{3} < 1$: inner loop.

b $r = 3 + 2\cos\theta$ $1 < \frac{a}{b} = \frac{3}{2} < 2$: dimpled.

c $r = 4 + \cos\theta$ $\frac{a}{b} = 4 > 2$: convex (no dimple).

(a) $\frac{a}{b} = \frac{1}{3} < 1$: inner loop.; (b) $1 < \frac{a}{b} = \frac{3}{2} < 2$: dimpled.; (c) $\frac{a}{b} = 4 > 2$: convex (no dimple).

12 [Intermediate] • Target: Finding maximum and minimum r-values

For the limaçon $r = 3 + 2\sin\theta$, find the maximum and minimum values of r, and the angles at which they occur.

Maximum $r = 5$ at $\sin\theta = 1$ ($\theta = \frac{\pi}{2}$). Minimum $r = 1$ at $\sin\theta = -1$ ($\theta = \frac{3\pi}{2}$).

13 [Intermediate] • Target: Comparing sine- and cosine-based polar curves

Explain how the graph of $r = 3 + 3\sin\theta$ compares to $r = 3 + 3\cos\theta$ (same cardioid shape, different orientation).

Both are cardioids of the same size and shape. The cosine version $r = 3 + 3\cos\theta$ is symmetric about the polar axis (horizontal), pointing right, while the sine version $r = 3 + 3\sin\theta$ is symmetric about $\theta = \frac{\pi}{2}$ (vertical), pointing up — a $90°$ rotation of one relative to the other.

14 [Intermediate] • Target: Sketching from key features

Without fully graphing, list the key features you would compute first to sketch $r = 2 - 4\cos\theta$ (max/min r, pole crossings, symmetry).

Type: limaçon with inner loop ($|a| = 2 < |b| = 4$). Symmetry: about the polar axis (cosine-based). Max $|r| = 6$ at $\theta = \pi$; min value $r = -2$ at $\theta = 0$. Pole crossings: $2 - 4\cos\theta = 0 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$ (marking where the inner loop begins/ends).

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: A capstone graphing problem** • **Cross-Lesson: 2.02 Radian Measure and Arc Length**

For the rose $r = 4\cos(3\theta)$, find: (a) the number of petals, (b) the maximum value of r, (c) the value(s) of $\theta \in [0, \pi)$ where the first petal reaches its maximum length, and (d) the angles where the curve passes through the pole in $[0, \pi)$.

(a) $n = 3$ is odd: 3 petals. (b) Maximum $|r| = 4$. (c) $\cos(3\theta) = 1 \Rightarrow 3\theta = 0 \Rightarrow \theta = 0$ (first petal's tip). (d) $\cos(3\theta) = 0 \Rightarrow 3\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$.

16 **[Challenging]** • **Target: Distinguishing curve families by equation form alone** • **Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions**

Without graphing, state whether each polar equation is a circle, a line, a rose, or a limaçon, based purely on its algebraic form: $r = 7$, $r = 5\sin(4\theta)$, $\theta = \frac{2\pi}{5}$, $r = 2 - 2\cos\theta$.

$r = 7$: circle (constant r). $r = 5\sin(4\theta)$: rose ($n = 4$ even, 8 petals). $\theta = \frac{2\pi}{5}$: line through the pole (constant θ). $r = 2 - 2\cos\theta$: cardioid (a special limaçon, coefficient equals constant).

17 **[Challenging]** • **Target: Comparing sine- and cosine-based polar curves** • **Cross-Lesson: 2.04 Law of Sines and Law of Cosines**

The curve $y = x^2 - 6x + 5$ can be mapped to $y = x^2$ by a sequence of two geometric transformations. Express the equation in vertex form, state the vertex, and describe fully each transformation.

Complete the square: $y = (x - 3)^2 - 9 + 5 = (x - 3)^2 - 4$. The vertex of the parabola is $(3, -4)$. To map $y = (x - 3)^2 - 4$ back to $y = x^2$, we must translate the graph 3 units to the left and 4 units upwards. In vector notation, this is a single translation by $(-3; 4)$, or two sequential transformations: (1) Translation by vector $(-3; 0)$ (3 units left), and (2) Translation by vector $(0; 4)$ (4 units up).

18 **[Challenging]** • **Target: Sketching from key features** • **Cross-Lesson: 3.01 The Unit Circle and Exact Values**

Convert the polar equation $r = 6\cos\theta - 8\sin\theta$ to Cartesian rectangular form, and identify the geometric curve, its center, and its radius.

Step 1: Multiply both sides by r: $r^2 = 6r\cos\theta - 8r\sin\theta$. Step 2: Substitute $r^2 = x^2 + y^2$, $r\cos\theta = x$, and $r\sin\theta = y$: $x^2 + y^2 = 6x - 8y \Rightarrow (x^2 - 6x + 9) + (y^2 + 8y + 16) = 9 + 16 = 25$. $\mathbf{(x - 3)^2 + (y + 4)^2 = 25}$. The curve is a circle with center $\mathbf{(3, -4)}$ and radius $\mathbf{R = 5}$.

19 **[Challenging]** • **Target: A capstone graphing problem** • **Cross-Lesson: 3.02 Fundamental Trigonometric Identities**

Analyze the polar curve $r(\theta) = 2 - 4\cos\theta$ for $0 \le \theta < 2\pi$. Identify the curve type, find the maximum distance from the pole, and determine the exact angles θ where the curve passes through the pole.

Step 1: Classify the limaçon $r = a - b\cos\theta$: Since $\frac{a}{b} = \frac{2}{4} = \frac{1}{2} < 1$, the curve is a limaçon with an inner loop. Step 2: Maximum distance from the pole: Occurs when $\cos\theta = -1$ ($\theta = \pi$): $r_{max} = 2 - 4(-1) = \mathbf{6}$. Step 3: Angles at the pole ($r = 0$): $2 - 4\cos\theta = 0 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \boldsymbol{\theta = \frac{\pi}{3}, \frac{5\pi}{3}}$.

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20 [Challenging] • Target: Distinguishing curve families by equation form alone • Cross-Lesson: 3.03 Sum and Difference Formulas

Find the set of values of the constant k for which the equation $kx^2 + 4x + (k - 3) = 0$ has real roots.

For real roots, the discriminant $\Delta = b^2 - 4ac \geq 0$. Here $a = k$, $b = 4$, $c = k - 3$: $\Delta = 4^2 - 4(k)(k - 3) = 16 - 4k^2 + 12k \geq 0$ Divide the entire inequality by -4, reversing the inequality sign: $k^2 - 3k - 4 \leq 0$ Factor the quadratic: $(k - 4)(k + 1) \leq 0$. The critical values are $k = -1$ and $k = 4$. Since the parabola opens upward, the product is ≤ 0 between the roots: $-1 \leq k \leq 4$. Note that if $k = 0$, the equation becomes the linear equation $4x - 3 = 0$, which has one r.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Recognizing polar graphs

The graph of the polynomial function $f(x) = x^3 + ax^2 + bx - 6$ has a local maximum at $x = -3$ and a local minimum at $x = 1$. Find the values of a and b.

a. $a = -3, b = 9$

b. $a = 3, b = -9$ ✓

c. $a = -6, b = -3$

d. $a = 6, b = 3$

For a polynomial function to have local extrema at specific x-values, its derivative must be zero at those points. The derivative of $f(x) = x^3 + ax^2 + bx - 6$ is $f'(x) = 3x^2 + 2ax + b$. Since there are local extrema at $x = -3$ and $x = 1$, we have $f'(-3) = 0$ and $f'(1) = 0$. Substituting $x = -3$ into $f'(x)$: $3(-3)^2 + 2a(-3) + b = 0$ $3(9) - 6a + b = 0$ $27 - 6a + b = 0$ (Equation 1) Substituting $x = 1$ into $f'(x)$: $3(1)^2 + 2a(1) + b = 0$ $3 + 2a + b = 0$ (Equation 2) Now we have a system of two line.

22 [Foundational] • Target: Symmetry tests

The minimum point of $y = x^2$ is translated by the vector $(-3; 6)$ to give curve C. State the coordinates of the minimum point of C.

a. (3,6)

b. (6,-3)

c. **(-3,6)** ✓

d. (-3,-6)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-3, 6)$.

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23 [Foundational] • Target: Graphing from a table of values

The graph of the function $f(x) = -(x-3)^2 + 4$ is shown in the xy-plane below. What is the maximum value of $f(x)$?

a. -5

b. 3

c. 4 ✓

d. 7

The maximum value of a function refers to the greatest output value (or y-value) that the function attains across its domain. The given quadratic function is written in vertex form, $f(x) = a(x-h)^2 + k$, where $a = -1$, $h = 3$, and $k = 4$. 1. For any real number x, the squared term is non-negative: $(x-3)^2 \geq 0$. 2. Multiplying by -1 reverses the inequality: $-(x-3)^2 \leq 0$. 3. Adding 4 to both sides gives: $-(x-3)^2 + 4 \leq 4$, which means $f(x) \leq 4$ for all x. When $x = 3$, the funct.

24 [Foundational] • Target: Rose curves in depth

The complex number z satisfies the equation $|z - (6+8\mathrm{i})| = 4$. Given that z^* denotes the complex conjugate of z, determine the maximum possible value of $|z - \mathrm{i}z^*|$.

a. $8 + 2\sqrt{2}$ ✓

b. $8 - 2\sqrt{2}$

c. $4\sqrt{2} + 2$

d. $4 + \sqrt{2}$

Let $z = x + \mathrm{i}y$, where $x, y \in \mathbb{R}$. Then $z^* = x - \mathrm{i}y$. First, express $z - \mathrm{i}z^*$ in Cartesian components: $\mathrm{i}z^* = \mathrm{i}(x - \mathrm{i}y) = y + \mathrm{i}x$, $z - \mathrm{i}z^* = (x + \mathrm{i}y) - (y + \mathrm{i}x) = (x-y) + \mathrm{i}(y-x) = (x-y)(1-\mathrm{i})$. Taking the modulus on both sides: $|z - \mathrm{i}z^*| = |x-y| \cdot |1-\mathrm{i}| = \sqrt{1^2 + (-1)^2}\,|x-y| = \sqrt{2}\,|x-y|$. The given condition $|z - (6+8\mathrm{i})| = 4$ d.

25 [Foundational] • Target: Identifying limaçon subtypes

Consider the rational equation $\frac{x+1}{x-3} + \frac{k}{x+1} = \frac{x+13}{x^2-2x-3}$ where k is a constant. What is the sum of all real values of k for which the equation has no real solution?

a. -10 ✓

b. -7

c. -3

d. 10

Step 1: Determine the restricted values from the denominators. The denominators are $x-3$, $x+1$, and $x^2 - 2x - 3 = (x-3)(x+1)$. Thus, the domain of the equation is all real numbers except $x = 3$ and $x = -1$. Step 2: Clear the denominators by multiplying both sides by $(x-3)(x+1)$: $(x+1)^2 + k(x-3) = x + 13$ Step 3: Expand and rewrite as a standard quadratic equation: $(x^2 + 2x + 1) + kx - 3k = x + 13$ $x^2 + (k+1)x - (3k+12) = 0$ Step 4: Factor the quadratic equation. Notice that substitu.

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26 [Foundational] • Target: Finding maximum and minimum r-values

What are the values of x that satisfy the equation $|3x - 2| = 2|x + 1|$?

a. $x = 0$ **and** $x = 4$ ✓

b. $x = 0$ and $x = -4$

c. $x = \frac{1}{5}$ and $x = 4$

d. $x = \frac{1}{4}$ and $x = \frac{3}{2}$

To solve the modulus equation $|3x - 2| = 2|x + 1|$, we can either consider the linear cases or square both sides. **Method 1: Considering linear cases** The equation $|A| = |B|$ (where $B = 2(x + 1)$) is equivalent to $A = B$ or $A = -B$: - Case 1: $3x - 2 = 2(x + 1) \Rightarrow 3x - 2 = 2x + 2 \Rightarrow x = 4$ - Case 2: $3x - 2 = -2(x + 1) \Rightarrow 3x - 2 = -2x - 2 \Rightarrow 5x = 0 \Rightarrow x = 0$ **Method 2: Squaring both sides** Since both sides are non-negative, we can square both sides.

27 [Foundational] • Target: Comparing sine- and cosine-based polar curves

A factory produces two types of components, A and B, using two machines, M1 and M2. The time (in minutes) required to produce one unit of each component on each machine is given in the table below: | Component | Machine M1 | Machine M2 | |---|---|---| | A | 2 | 3 | | B | 4 | 1 | The factory has 480 minutes of M1 time and 240 minutes of M2 time available per day. Let x be the number of units of component A produced per day and y be the number of units of component B produced per day. Suppose the profit from selling one unit of component A is 5 and the profit from selling one unit of component B is 8. The factory wants to maximize its total daily profit. Formulate this problem as a linear programming problem, defining the objective function and the constraints.

a. **Maximize** $P = 5x + 8y$ **subject to** $2x + 4y \leq 480$**,** $3x + y \leq 240$**,** $x \geq 0$**,** $y \geq 0$**.** ✓

b. Maximize $P = 8x + 5y$ subject to $2x + 3y \leq 480$, $4x + y \leq 240$, $x \geq 0$, $y \geq 0$.

c. Maximize $P = 5x + 8y$ subject to $4x + 2y \leq 480$, $y + 3x \leq 240$, $x \geq 0$, $y \geq 0$.

d. Maximize $P = 5x + 8y$ subject to $2x + 4y \leq 240$, $3x + y \leq 480$, $x \geq 0$, $y \geq 0$.

The objective is to maximize the total daily profit. The profit from selling x units of component A is $5x$ and the profit from selling y units of component B is $8y$. Therefore, the objective function is $P = 5x + 8y$. The constraints are based on the available machine time. For Machine M1, producing x units of A takes $2x$ minutes and producing y units of B takes $4y$ minutes. The total time on M1 must be less than or equal to the available 480 minutes: $2x + 4y \leq 480$. For Machine M2, producing x un.

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28 [Foundational] • Target: Sketching from key features

The graph of the polynomial function $f(x) = x^3 - ax^2 + bx - 6$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. Determine the value of $a + b$.

a. 5

b. **-5 ✓**

c. 1

d. -1

To find the local extrema of a polynomial function, we need to find its critical points by taking the derivative and setting it to zero. The derivative of $f(x)$ is $f'(x) = 3x^2 - 2ax + b$. Since $f(x)$ has local extrema at $x = -1$ and $x = 3$, these are the roots of $f'(x) = 0$. Thus, we can write $f'(x)$ in factored form: $f'(x) = k(x - (-1))(x - 3) = k(x + 1)(x - 3)$ for some constant k. Expanding this, we get $f'(x) = k(x^2 - 2x - 3)$. Comparing the coefficient of x^2 in $f'(x) = 3x^2 - 2ax + b$ and.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A capstone graphing problem

A research team is studying the impact of different fertilizer types on crop yield. They conducted an experiment with two fertilizer types, A and B, and recorded the yield in kilograms per plot. The data is summarized in the two-way table below. The total number of plots is 200. | Fertilizer Type | Yield < 50 kg | Yield ≥ 50 kg | Total | |---|---|---|---| | A | 30 | 70 | 100 | | B | 50 | 50 | 100 | | Total | 80 | 120 | 200 | Given that a plot was selected at random and its yield was less than 50 kg, what is the probability that it received fertilizer type B? Provide your answer as a fraction in its simplest form.

a. 1/2

b. **5/8 ✓**

c. 3/8

d. 1/4

Let Y be the event that the yield is less than 50 kg, and B be the event that the fertilizer type is B. We are asked to find the conditional probability P(B|Y). From the table, we have: Number of plots with yield < 50 kg (event Y) = 80. Number of plots with yield < 50 kg AND fertilizer type B (event B ∩ Y) = 50. The formula for conditional probability is P(B|Y) = P(B ∩ Y) / P(Y). Alternatively, using the counts directly: P(B|Y) = (Number of plots with yield < 50 kg and fertilizer type B) / (Number of plot.

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30 [Intermediate] • Target: Distinguishing curve families by equation form alone

The graph of the cubic polynomial $P(x) = x^3 + ax^2 + bx + c$ has a local maximum at $(-1, 8)$ and a local minimum at $(3, -24)$. Find the value of a.

a. -3 ✓

b. -1

c. 1

d. 3

The derivative of the polynomial is $P'(x) = 3x^2 + 2ax + b$. Local extrema occur where $P'(x) = 0$. Thus, the roots of $P'(x)$ are $x = -1$ and $x = 3$. Using Vieta's formulas for the quadratic $P'(x)$, the sum of the roots is $-1 + 3 = 2$, and from the polynomial, the sum of the roots is $-\frac{2a}{3}$. Equating these, we get $2 = -\frac{2a}{3}$, which simplifies to $6 = -2a$, so $a = -3$. Alternatively, we can set up a system of equations. Since $x = -1$ and $x = 3$ are roots of $P'(x)$, we have.

31 [Intermediate] • Target: Recognizing polar graphs

Consider the function $g(x) = x^2 - ax + b$. If the graph of $y = g(x)$ has a minimum value of -1 at $x = 3$, find the values of a and b.

a. $a = 6, b = 8$ ✓

b. $a = -6, b = 8$

c. $a = 6, b = -1$

d. $a = -6, b = -1$

The function is a quadratic $g(x) = x^2 - ax + b$. The x-coordinate of the vertex (where the minimum occurs) is given by $-(-a)/(2 \cdot 1) = a/2$. We are given that the minimum occurs at $x = 3$. Therefore, $a/2 = 3$, which implies $a = 6$. The minimum value of the function is $g(3)$. Substituting $a = 6$ into the function, we get $g(x) = x^2 - 6x + b$. The minimum value is $g(3) = 3^2 - 6(3) + b = 9 - 18 + b = -9 + b$. We are given that the minimum value is -1. So, $-9 + b = -1$, which implies $b = 8$. Thus.

32 [Intermediate] • Target: Symmetry tests

Consider a cubic polynomial $P(x) = ax^3 + bx^2 + cx + d$. The graph of $P(x)$ has a y-intercept at $(0, 12)$. It is known that $P(x)$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. If the value of the local maximum is 20, determine the value of the constant term d of the polynomial.

a. **12** ✓

b. 10

c. 8

d. 6

Let the cubic polynomial be $P(x) = ax^3 + bx^2 + cx + d$. \\ We are given that the y-intercept is at $(0, 12)$, so $P(0) = 12$. \\ Substituting $x = 0$ into the polynomial, we get $P(0) = a(0)^3 + b(0)^2 + c(0) + d = d$. \\ Therefore, $d = 12$. \\ The problem statement asks for the value of the constant term d, which is directly given by the y-intercept.

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33 [Intermediate] • Target: Graphing from a table of values

Express $\frac{1+\cos\theta+i\sin\theta}{1+\cos\theta-i\sin\theta}$ in polar form.

a. $e^{i\theta}$ ✓

b. 1

c. $e^{2i\theta}$

d. $\cos\theta+i\sin\theta$ only for $\theta=0$

Numerator: $2\cos^2(\theta/2)+2i\sin(\theta/2)\cos(\theta/2)=2\cos(\theta/2)e^{i\theta/2}$. Denominator: $2\cos(\theta/2)e^{-i\theta/2}$. Ratio: $e^{i\theta}$.

34 [Intermediate] • Target: Rose curves in depth

Find the maximum value of $f(x)=x^a(1-x)^b$ on $[0,1]$ where $a,b>0$.

a. $\left(\frac{a}{a+b}\right)^a\left(\frac{b}{a+b}\right)^b$

b. $\frac{a^a b^b}{(a+b)^{a+b}}$

c. $\frac{1}{4}$ for $a=b=1$

d. **Both A and B** ✓

At maximum $x=a/(a+b)$: $f=\left(\frac{a}{a+b}\right)^a\left(\frac{b}{a+b}\right)^b$. This equals $\frac{a^a b^b}{(a+b)^{a+b}}$. Both A and B are the same.

35 [Intermediate] • Target: Identifying limaçon subtypes

If $z_1=2\angle 45°$ and $z_2=3\angle 30°$ (polar form), find $|z_1 z_2|$.

a. $6\angle 75°$

b. 5

c. $\sqrt{13}$

d. 6 ✓

Applying Quadratics & Complex Numbers principles, $|z_1z_2|=|z_1||z_2|=2\cdot 3=6$. Therefore, the verified correct option is 6.

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36 [Intermediate] • Target: Finding maximum and minimum r-values

Identify the Cartesian curve represented by $r = 6\cos\theta$.

a. Circle of radius 6 centered at $(0, 0)$

b. **Circle of radius 3 centered at $(3, 0)$** ✓

c. Cardioid with parameter $a = 3$

d. Ellipse with semi-major axis 6

Multiply by r: $r^2 = 6r\cos\theta \Rightarrow x^2 + y^2 - 6x = 0 \Rightarrow (x-3)^2 + y^2 = 9$. Center $(3, 0)$, radius 3.

37 [Intermediate] • Target: Comparing sine- and cosine-based polar curves

Find the principal cube root of $8i$ in polar form.

a. $2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

b. $8\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

c. $2(\cos 0 + i\sin 0)$

d. $2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$ ✓

$8i = 8\text{cis}(\pi/2)$. Principal cube root: $\sqrt[3]{8}\,\text{cis}\left(\frac{\pi/2}{3}\right) = 2\text{cis}(\pi/6)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Sketching from key features • Cross-Lesson: 3.01 The Unit Circle and Exact Values

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-2;\ -6)$. State the coordinates of the new vertex.

a. (-2,6)

b. **(-2,-6)** ✓

c. (-6,-2)

d. (2,-6)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-2,\ -6)$.

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39 [Challenging] • Target: A capstone graphing problem • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

The curve $y = x^2$ is translated by the vector $(7; 4)$. State the coordinates of the new vertex.

a. (-7,4)

b. **(7,4)** ✓

c. (4,7)

d. (7,-4)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(7, 4)$.

40 [Challenging] • Target: Distinguishing curve families by equation form alone • Cross-Lesson: 3.03 Sum and Difference Formulas

Identify the polar curve $r = 5 + 3\cos\theta$.

a. a cardioid

b. a circle

c. **a dimpled limaçon** ✓

d. a rose with 3 petals

The equation $r = 5 + 3\cos\theta$ has the standard polar form matching: a dimpled limaçon.

41 [Challenging] • Target: Recognizing polar graphs • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

The point $(3, -2)$ lies on the curve $y = f(x)$. Find the coordinates of the corresponding point on $y = f(x + 5) - 2$.

a. **(-2,-4)** ✓

b. (-2,4)

c. (3,-2)

d. (-4,-2)

Since replace x by x-(-5) and add -2 to the output, the image point is $(-2, -4)$.

42 [Challenging] • Target: Symmetry tests • Cross-Lesson: 4.01 Circles and Parabolas

Convert $r = \frac{4}{1 - 2\cos\theta}$ to Cartesian form and identify its type.

a. Parabola: $y^2 = 4(2)x$

b. **Hyperbola:** $3x^2 - y^2 + 16x + 16 = 0$ ✓

c. Circle: $x^2 + y^2 = 4$

d. Ellipse: $3x^2 + y^2 = 16$

$e = 2 > 1$, so it is a hyperbola. In Cartesian: $3x^2 - y^2 + 16x + 16 = 0$.

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43 [Challenging] • Target: Graphing from a table of values • Cross-Lesson: 4.02 Ellipses

For what values of x does x^x reach a minimum?

a. $x = 0$

b. $x = e$

c. $x = 1$

d. $x = 1/e$ ✓

$f(x) = x^x$. $f'(x) = x^x(1 + \ln x) = 0 \Rightarrow \ln x = -1 \Rightarrow x = 1/e$.

44 [Challenging] • Target: Rose curves in depth • Cross-Lesson: 4.03 Hyperbolas

Find the equation of the line through $(-1, -4)$ with gradient -2, in the form $y = mx + c$.

a. y=2x-6

b. y=-2x-5

c. **y=-2x-6** ✓

d. y=-2x+6

Using $y - y_1 = m(x - x_1)$: $y + 4 = -2(x + 1)$, which rearranges to $y = -2x - 6$.

45 [Challenging] • Target: Identifying limaçon subtypes • Cross-Lesson: 5.01 Vectors: Components and Operations

Determine the type and key features of $16x^2 - 9y^2 = 144$.

a. Parabola

b. Ellipse

c. **Hyperbola:** $x^2/9 - y^2/16 = 1$, $a = 3, b = 4, c = 5$ ✓

d. Circle

Divide by 144: $x^2/9 - y^2/16 = 1$. $c = \sqrt{9 + 16} = 5$.

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